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COMPREHENSIBILITY OF PERSUASIVE ARGUMENTS AS A DETERMINANT OF OPINION CHANGE [1]

ALICE H. EAGLY [2]

University of Massachusetts, Amherst

In three experiments involving two different manipulations of the comprehensibility of persuasive arguments, lowering comprehensibility lessened acceptance of the conclusion of a message. Since argument comprehensibility did not interact with discrepancy size, communicator credibility, or counterarguing, this finding proved extremely general. Explanation focused on two processes that accompanied the deterioration of comprehensibility: (*a*) the lesser amount of supportive information that was received and (*b*) the more negative affective reaction that was produced. Experiment 3 assessed the two explanations and concluded that both contributed to the message comprehensibility effects.

The proposition that the psychological processes involved in the reception of a persuasive message mediate opinion change has often been stated. Attention to the message and comprehension of it were important antecedent conditions of opinion change in the writings of the Yale attitude-change researchers (e.g., Hovland, Janis, & Kelley, 1953). This approach has been emphasized more recently by McGuire (1968, 1969), especially in his model of the relationship between personality variables and influenceability. Yet, despite this emphasis on reception processes, there is little research that has directly examined their relationship to persuasion.

Existing research on this issue is almost exclusively correlational. It does not involve manipulations of the reception process itself but of other variables (e.g., communicator credibility) that may influence both reception and acceptance of the message (see reviews by McGuire, 1966, 1969). In general, the findings in this area fail to yield consistent evidence of a positive relationship between retention of message content and attitude change (e.g., Hovland & Weiss, 1952; Millman, 1968; Osterhouse & Brock, 1970; Thistlethwaite, deHaan, & Kamenetsky, 1955; Zimbardo & Ebbesen, 1970), although some positive findings have been obtained (e.g., Haaland & Venkatesan, 1968; Janis & Rife, 1959; McGuire, 1957; Miller & Campbell, 1959).

In an effort to manipulate reception directly, Cook (1969) varied the number of arguments supporting a conclusion and obtained evidence (in one of two experiments reported) of greater attitude change toward the message supported by the greater number of arguments. Yet Cook's manipulation seems somewhat unrelated to the reception variable as it is conceptualized here, since the difference between experimental conditions was a function of the absence of parts of a message rather than of omissions in or defects in the reception process itself. In addition, since the reception of a complex persuasive message no doubt improves with repetition, it is of some interest to note that Weiss (1968) has found persuasion (when measured by speed of agreement) to be an increasing function of the number of times the subject is exposed to a communication.

The fact that findings concerning the relationship between reception and attitude change present a mixed pattern has been used by some (e.g., Fishbein & Ajzen, 1972; Greenwald, 1968) to suggest that reception does not mediate attitude change or is at

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[2] Requests for reprints should be sent to Alice H. Eagly, Department of Psychology, University of Massachusetts, Amherst, Massachusetts 01002.

least rather unimportant in the attitude-change process. However, such views may represent overly strong interpretations of the failure to obtain consistently positive relationships between verbal memory and attitude change.

A major problem in evaluating most existing evidence on the relationship between reception and persuasion stems from the reliance on verbal memory as the primary route to understanding reception. This procedure is probably misleading since reception refers to the process of encoding a message, while verbal memory encompasses encoding the message, storing it in memory, and then subsequently decoding it. That the decoded message may misrepresent reception becomes apparent when persuasion is conceptualized as the process of extracting meaning from aspects of the persuasive message as it is received, of accepting or yielding to this information, and of combining it into an overall changed attitude or opinion. After this processing has taken place, memory for the communication may no longer be necessary to the persuasion process. Consequently, verbal memory may obey rules different from those of attitude or opinion change (Anderson & Hubert, 1963; Insko, 1964; Watts & McGuire, 1964). Thus, verbal memory for the communication and attitude change are best conceptualized as different systems that may be functionally autonomous to some degree. Recognition and recall measures then provide a weak base on which to build an evaluation of the mediational role of message reception. To avoid exclusive reliance on measures of retention and to clarify the causal significance of reception, the present series of experiments employed experimental manipulations of reception.

There is yet another major problem that should be taken into account in interpreting most of the existing evidence concerning reception and persuasion. This problem is that the typical retention measures used in these studies (e.g., Thistlethwaite et al., 1955) combine into a single index memory for aspects of the message likely to show differing relationships to attitude change. In particular, past research generally fails to distinguish between the position or conclusion advocated by the message and the arguments supporting the position. Reception of these two aspects of the message may affect the persuasive process in differing ways, and distinctions between still other aspects of comprehension have been suggested by Watts and McGuire (1964).

The present research neglects the relationship between persuasion and reception of the position advocated in favor of focusing on reception of persuasive arguments. Good reception of the arguments supporting the conclusion of a message is expected to facilitate attitude change, since arguments generally provide some of the principal cues inducing acceptance of the advocated position. Despite the common-sense flavor of this hypothesis, it might be noted that there do exist viewpoints implying that the relationship between reception of arguments and persuasion may not be strong. For example, conceptualizations of attitude change such as prestige suggestion (e.g., Asch, 1952, pp. 387–449) or McGuire's (1969) "lazy organism" message recipient imply that the individual is affected mainly by the more global cues of the message's position and the identity of its source rather than by persuasive arguments. Another possibility is that reception of the content of the arguments may be rather unimportant because it may be more crucial to know that arguments exist than to understand their content. In contrast to these ideas, the hypothesis of a positive relationship between reception of arguments and persuasion rests on the assumption of a rather rational and content-oriented (e.g., Kelman & Eagly, 1965) message recipient who accepts a conclusion on the basis of his evaluation of arguments that favor it.

Even if the reception–persuasion relationship is, in general, positive, certain conditions may weaken the relationship. For example, for the relationship to be positive, the arguments themselves, when comprehended, must be accepted, and they must be perceived as evidence leading to the conclusion of the message (see Fishbein & Azjen's, 1972, discussion of the role of persuasive arguments in persuasion). In addition, Cook (1969) suggested that a positive relationship between reception and persuasion is more likely for

attitude formation than for attitude change. This idea follows from the assumption that the impact of persuasive arguments is weaker to the extent that they must compete with information that the message recipient already has concerning the issue.

The three experiments reported here manipulated message reception by varying the comprehensibility of persuasive arguments. In addition, the size of the discrepancy between the recipient's position and the position advocated was manipulated. Experiment 2 also manipulated the subject's level of counterarguing, and Experiment 3 manipulated communicator credibility. Inclusion of these additional variables allowed the exploration of a few conditions that might limit the expected positive relationship between message comprehensibility and persuasion. These additional variables were chosen according to the straightforward logic that any factor weakening the persuasiveness of the message might, as a consequence, weaken the relationship between argument comprehensibility and persuasion. In other words, good comprehension of persuasive arguments may do little to enhance acceptance of a recommendation that would not be accepted very strongly in any case, for example, one that is very discrepant or that is counterargued strongly by the recipient or one that is attributed to a source of unacceptably low credibility.

Experiment 1 employed a 2 × 3 factorial design with two levels of comprehensibility and three levels of discrepancy size.

EXPERIMENT 1

Method

Subjects

The subjects were 291 students (141 males and 150 females) enrolled in an introductory psychology course at the University of Illinois that required participation in experiments. Two subjects were eliminated because of incomplete responding.

Procedure

For convenience, subjects were scheduled in groups of six but were separated by partitions so that no communication could take place between them.

To justify presentation of the persuasive message, the experimenter explained that the research concerned how students evaluate college lecturers. He mentioned that students from introductory psychology classes were being asked to listen to short lectures by different professors from the Psychology Department of the University of Illinois and that professors were chosen who were willing to talk about controversial topics.

In order to justify the introduction of a written passage designed to manipulate availability of counterarguments, the experimenter said that a factor determining how students react to lectures is "how much the student already knows about the lecture before coming to class" and that half of the subjects would read an article relevant to the lecture that they would hear, while the other half would read an article pertaining to one of the other lectures.

The experimenter explained that another factor that may determine a student's reaction to a lecturer is how well he understands the lecture itself. The subject was told that in order to simulate different levels of understanding, some subjects would listen to a tape technically produced to be difficult to understand, while others would listen to a good-quality tape.

The relevant or irrelevant reading material was then distributed, and subjects were given 10 minutes to read it. The relevant article concerned sleep research and had embedded in it several arguments countering the arguments used in the persuasive communication that the subjects would hear. The irrelevant article concerned intelligence. Since this manipulation, designed to influence counterarguing, did not significantly affect any response, it is not considered further. The data were pooled over these conditions.

The experimenter then announced that the lecture would deal with sleep research and would be given by "Dr. Robert Paulsen, who is a physiological psychologist in this department" engaged in sleep research. The subjects were further told that Dr. Paulsen would sum up his discussion "with a recommendation to the average adult that you may find fairly surprising, namely, that the average person should get only _____ hours of sleep a day. However, he goes on to explain that this sleep must be of a certain kind and done in a certain way." The blank in the above passage was filled in with either six, four, or two hours, depending on the experimental condition to which the subject was assigned. The subjects were then asked to listen to the tape as they would to an "average college lecture" and were cautioned not to stop listening just because they might have gotten a bad tape. Immediately after the message, the subjects responded to a questionnaire designed to assess the extent of their agreement with the communication as well as other responses.

Control subjects were exposed to identical procedures until the point at which the tapes would be played. At that point, the experimenter introduced a questionnaire with a single item assessing the subject's belief concerning the number of hours of sleep that is desirable.

All of the subjects were debriefed orally and were later sent a written report of the findings of the experiment.

Persuasive Communication

The 11-minute message began with introductory statements that mentioned the overall recommendation (six, four, or two hours) and continued with six clearly enumerated arguments in support of the recommendation. Sample arguments are the following: (*a*) The amount of sleep that people get varies between cultures and is thus arbitrary. (*b*) Only rapid eye movement sleep really rests people, and therefore other phases of sleep can largely be dispensed with. (*c*) People can be trained to get mostly rapid eye movement sleep through a program that involves taking three or four evenly spaced naps per day.

The comprehensibility manipulation was accomplished by playing to half of the subjects a good-quality recording and to the other half, a poor-quality recording, which had been produced by overrecording and the introduction of background noise on the tape.

Measuring Instruments

Acceptance of the communicator's recommendation. On 7-point bipolar rating scales, subjects indicated how much they agreed with the recommendation given in the lecture and rated the convincingness of the arguments that the lecturer gave for his recommendation. They also indicated how many hours of sleep per 24 hours they believed desirable for the average adult for "maximum happiness, well-being, and success in life." The two acceptance measures, agreement and number of hours of sleep, correlated −.32 (the direction is negative because acceptance was indicated by a high score on the agreement scale but by preference for a low number of hours of sleep).

Counterarguing. Following a procedure described by Osterhouse and Brock (1970), subjects were given three minutes to "List below your thoughts and ideas about the possible effects of reducing amount of sleep to the level suggested in the lecture." In order to determine the number of counterarguments used, subjects' statements were content analyzed according to procedures adapted from Brock (1967) and Osterhouse and Brock (1970). A statement was scored as a counterargument if it indicated a negative consequence of the recommendation or constituted a logical attack on some aspect of the message. Negative consequences stated in the form of a question (e.g., "Wouldn't a person get run down after a few months of getting so little sleep?") were also scored as counterarguments. The counterargument scorings of the two raters correlated .94. Unelaborated rejection statements (e.g., "It's nonsense") were scored separately, but since their frequency was very low and interrater agreement was poor, these data were not reported.

Evaluation of source and message. Using 7-point bipolar rating scales, a subject evaluated (on well-informed, competent, unbiased, credible, and fair) the credibility of the source (i.e., the lecturer) and then evaluated (on reasonable, logical, well-written, good job, and sensible) the message itself.

Retention. In order to assess retention of the position the message advocated, subjects were asked to state the number of hours of sleep that the lecturer recommended. Retention of the persuasive arguments was tested by both recall and recognition. Subjects were first asked to restate the six main arguments contained in the message. These responses were scored for correctness by two raters (whose scorings correlated .94). Seventeen multiple-choice items tested retention of specific points mentioned in the arguments. The correlation between the two retention measures was .58.

Initial willingness to accept the recommended position. As a check on the subjects' perception of the discrepancy manipulation, they were asked to indicate on 7-point rating scales their reactions to four questions phrased in the following way: "When the person in charge of today's session told you that the lecturer would recommend the number of hours of sleep that he did, how . . ." The questions that followed inquired about the subject's willingness to go along with the source's views, his faith that the source's views would be sound and based on valid scientific evidence, his distrustfulness of the correctness of the source's point of view about sleep, and the applicability of the source's point of view concerning sleep to the subject's own sleeping behavior.

Other measures. Also in the final section of the questionnaire, a subject indicated on 7-point bipolar rating scales how much effort he put into listening to the tape recording, how pleasant the experience of listening was, how distracting the recording was, and how difficult it was to hear the lecture. Finally, each subject was asked to give his own interpretation of the experiment and to indicate at what point he came up with this interpretation. Since coding of these answers indicated only mild levels of suspicion and insight into the design, no subject was removed on this basis (from any of the three experiments reported here).

Results

Manipulation Checks

Message comprehensibility had a strong effect on retention of the message: Good-comprehensibility subjects recalled 3.59 of the 6 arguments correctly, while poor-comprehensibility subjects recalled 1.76 ($F = 133.08$, $df = 1/247$, $p < .001$). Good-comprehensibility subjects answered 12.52 of the 17 multiple-choice items correctly, while poor-comprehensibility subjects answered 7.45 correctly ($F = 260.37$, $df = 1/247$, $p < .001$). Also, poor-comprehensibility subjects rated the lecture as more difficult to hear clearly (for good-comprehensibility subjects, $\bar{X} = 5.74$; for poor-comprehensibility subjects, $\bar{X} = 1.11$, where 7 signifies very easy and 1,

TABLE 1

AGREEMENT WITH THE COMMUNICATOR'S RECOMMENDATION: EXPERIMENT I

Discrepancy	Good comprehensibility	Poor comprehensibility
Small[a]	5.80 (41)	4.28 (43)
Medium[b]	4.55 (44)	3.57 (42)
Large[c]	4.00 (42)	3.27 (41)
$\bar{X}$	4.77	3.71

Note. Ratings are on a 7-point scale with 7 indicating strong agreement and 1, strong disagreement. Cell *n*s are given in parentheses.
[a] 6-hour recommendation.
[b] 4-hour recommendation.
[c] 2-hour recommendation.

very difficult; $F = 1316.07$, $df = 1/247$, $p < .001$).

Most of the subjects correctly recalled the communicator's recommendation, since it was stated by the experimenter as well as in the message. However, of the 13 subjects who did not state the correct number of hours, 12 were in the poor-comprehensibility condition. These subjects were retained in the analysis because the wording of the recall question may have led some subjects to believe that it was legitimate to report only what they had heard in the lecture, which was, of course, less comprehensible in the poor-comprehensibility condition.

The perceived acceptability of the communicator's recommendation varied with its extremity. As the recommendation became more extreme, subjects reported less initial willingness to accept it ($F = 13.87$, $df = 2/247$, $p < .001$), and in addition, counterarguing increased ($F = 4.97$, $df = 2/247$, $p < .001$).

Acceptance of the Communicator's Recommendation

As shown in Table 1, reported agreement with the communicator's recommendation increased as comprehensibility improved ($F = 38.85$, $df = 1/247$, $p < .001$) and decreased as discrepancy size increased ($F = 23.28$, $df = 2/247, p < .001$). Comprehensibility and discrepancy did not interact significantly ($F = 1.85$, $df = 2/247$). This negative relationship between discrepancy size and agreement is not necessarily inconsistent with the positive relationships usually reported between discrepancy size and opinion change (e.g., Bochner & Insko, 1966; Hovland & Pritzker, 1957), since the "after-only" agreement measure in the present study did not remove the agreement differences between discrepancy conditions that existed prior to receipt of the message.

Comprehensibility also affected the number of hours of sleep considered desirable for the average adult: Fewer hours were believed desirable in the good-comprehensibility condition ($\bar{X} = 6.98$) than in the poor-comprehensibility condition ($\bar{X} = 7.44$, $F = 12.35$, $df = 1/247$, $p < .001$). Hours of sleep was not affected by the extremity of the recommendation ($F = 1.83$, $df = 2/247$), nor was there a Comprehensibility × Discrepancy interaction ($F < 1$, $df = 2/247$). Also, the ideal number of hours of sleep estimated by the control group subjects, who were not exposed to the message or the recommended position, was 7.99, which was significantly different from the experimental group mean of 7.21 ($t = 4.26$, $df = 287$, $p < .001$). In addition, the rated convincingness of the arguments decreased as comprehensibility deteriorated ($p < .001$) but was not affected by discrepancy size, nor was there a significant interaction.

Other Measures

Both the source and the message were evaluated significantly more unfavorably in the poor-comprehensibility condition. Although evaluations of both the source and the message tended to become more negative as discrepancy size increased, the findings were not statistically significant.

The experience of listening to the tape was judged significantly less pleasant in the poor-comprehensibility condition, and the poor-comprehensibility message was rated significantly more distracting. Subjects' ratings of the effort they put into listening to the tape recording were not affected by any of the experimental variables, nor was the number of statements written in response to the open-ended question used to elicit counterarguments.

EXPERIMENT 2

In order to clarify further the relationship between comprehension and attitude change,

a second experiment was carried out. A different (and successful) counterarguing manipulation was employed. Exploration of counterargument production is important in order to evaluate whether good comprehensibility of persuasive arguments enhances the persuasiveness of a message even when subjects are motivated to counterargue what they comprehend. Also, since the comprehensibility manipulation in the first study had produced a rather severe decrement of comprehension, a third level of comprehensibility was added between the levels of the first study. The middle level of discrepancy (four hours) was dropped. Thus this study used a 3 × 2 × 2 factorial design with three levels of comprehensibility, two levels of counterarguing, and two levels of discrepancy size.

Method

Subjects

The subjects were 194 students (95 males and 99 females) enrolled in an introductory psychology course at the University of Illinois that required participation in experiments. One subject was eliminated because of incomplete responding. Forty-five additional subjects from a social psychology class served as an informal control group.

Procedure

The procedure, with the exception of the counterarguing manipulation, was the same as that of Experiment 1. Accordingly, after explaining that some tapes would be difficult to understand, the experimenter mentioned, "Another variable is the mood or frame of mind a listener is in when he comes to a lecture. I will give you written instructions explaining the frame of mind or role you should put yourself in." The written instructions then preceded the experimenter's introduction to the message.

High counterarguing subjects received the following statement:

> Your job is to take the role of an extremely critical student who raises many objections to the lecturer's points. In other words, while you are listening to the lecture, think of *all possible objections* to what the lecturer says and to his overall viewpoint. You will keep these objections to yourself during the lecture, but we will give you a test later on to find out how well you were able to play the role that you have been assigned. (This test will be the last thing you do in the experiment and is not part of the questionnaire that you will fill out immediately following the lecture.) Keep these instructions on your desk to remind you of what you are supposed to do while listening to the lecture.

Further down on the page was the notation: "REMEMBER—THINK OF AS MANY OBJECTIONS AS YOU CAN."

For low-counterarguing subjects, the first two sentences of the statement were as follows: "Your job is to take the role of a student who is listening to a lecture. In other words, listen to what the lecturer says in exactly the way that you would ordinarily do this in a course lecture." Furthermore, the notation further down on the page was the following: "REMEMBER—LISTEN AS YOU WOULD IN A COURSE LECTURE."

Persuasive Communication

The same persuasive communication was employed, though only in the six- and the two-hour versions. A middle level of comprehensibility was introduced, a tape with some distortion and background noise.

Measuring Instruments

The postquestionnaire was changed in a couple of minor respects. The item about the ideal number of hours of sleep was worded in terms of the number of hours that "you think you should get" instead of in terms of how much the average adult should get. Also, since the item *authoritative* in the communicator credibility set of items of Experiment 1 did not cohere very well with the other items, the term *unbiased* was substituted.

The item assessing recall of the communicator's position was worded "How many hours of sleep per 24 hours did the experimenter announce that the lecturer would recommend for the average person?"

The two raters' counterargument scorings correlated .93, while their scorings of the number of arguments recalled correlated .97. The two measures of acceptance of the communicator's recommendation correlated −.34, while the two retention measures correlated .57.

Results

Manipulation Checks

The mean number of counterarguments was 2.24 in the high-counterarguing condition and 1.71 in the low-counterarguing condition ($F = 6.21$, $df = 1/181$, $p < .05$). As in Experiment 1, the comprehensibility manipulation had a strong effect on comprehension of the message: Good-comprehensibility subjects recalled 4.08 of the 6 arguments correctly, medium-comprehensibility subjects recalled 3.07, and poor-comprehensibility subjects recalled 1.55 ($F = 71.74$, $df = 2/181$, $p < .001$). Good-comprehensibility subjects answered 12.84 of the 17 multiple-choice items

TABLE 2

AGREEMENT WITH THE COMMUNICATOR'S RECOMMENDATION: EXPERIMENT II

Discrepancy and counterarguing	Good comprehensibility	Medium comprehensibility	Poor comprehensibility
Small[a]			
Low	5.41 (17)	5.21 (14)	4.27 (15)
High	4.72 (18)	3.69 (16)	3.56 (18)
Large[b]			
Low	3.94 (16)	3.17 (12)	3.33 (18)
High	2.92 (13)	2.89 (19)	2.41 (17)
$\bar{X}$	4.34	3.69	3.37

Note. Ratings are on a 7-point scale, with 7 indicating strong agreement and 1, strong disagreement. Cell *ns* are given in parentheses.
[a] 6-hour recommendation.
[b] 2-hour recommendation.

correctly, medium-comprehensibility subjects answered 8.90, and poor-comprehensibility subjects answered 6.74 ($F = 135.29$, $df = 2/181$, $p < .001$). Also, ratings of the difficulty of hearing the lecture varied with comprehensibility (for good-comprehensibility subjects, $\bar{X} = 5.33$; for medium-comprehensibility subjects, $\bar{X} = 1.34$; for poor-comprehensibility subjects, $\bar{X} = 1.08$, where 7 signified very easy and 1, very difficult; $F = 349.18$, $df = 2/181$, $p < .001$). Fifteen subjects, who were retained in the analysis, failed to recall the recommended number of hours of sleep correctly, but they were quite evenly distributed across the experimental conditions.

As the recommendation became more extreme, subjects reported less initial willingness to accept it ($p < .001$), and the message elicited more counterarguing ($p < .01$).

Acceptance of the Communicator's Recommendation

As shown in Table 2, reported agreement with the communicator's recommendation increased as comprehensibility improved ($F = 6.02$, $df = 2/181$, $p < .01$). Agreement decreased as counterarguing increased ($F = 17.82$, $df = 1/181$, $p < .001$) and as discrepancy size increased ($F = 45.33$, $df = 1/181$, $p < .001$). Tukey (b) tests (Winer, 1962) of the difference in agreement between the good- and poor-comprehensibility conditions and the difference between the good- and medium-comprehensibility conditions were both significant. Also, as the message became more comprehensible, subjects believed that fewer hours of sleep were ideal for themselves (for good-comprehensibility subjects, $\bar{X} = 7.15$; for medium-comprehensibility subjects, $\bar{X} = 7.50$; for poor-comprehensibility subjects, $\bar{X} = 7.55$; $F = 3.08$, $df = 2/181$, $p < .05$). Neither counterarguing nor discrepancy affected estimates of the ideal number of hours of sleep, nor were there any interactions. The ideal number of hours of sleep estimated by the control group subjects, who were not exposed to the recommended position, was 8.13, which was significantly different from the experimental group mean of 7.40 ($t = 4.45$, $df = 236$, $p < .001$). Also, the rated convincingness of the arguments increased as comprehensibility improved ($p < .001$), decreased as discrepancy size increased ($p < .001$), and decreased as counterarguing increased ($p < .001$).

Other Measures

Both the source and the message were evaluated significantly more unfavorably as comprehensibility deteriorated, as counterarguing increased, and as discrepancy increased.

The rated pleasantness of listening to the tape increased significantly as comprehensibility improved, and ratings of the distractingness of the message had a significant indirect relationship with comprehensibility. Subjects' ratings of the effort they put into listening to the tape recording were not affected by any of the experimental variables, nor was the number of statements written in response to the open-ended question used to elicit counterarguing.

Discussion

The principal finding of Experiments 1 and 2 is that lowering the comprehensibility of the persuasive arguments of a message lessens acceptance of its conclusion. Not only did the persuasiveness of the good- and poor-comprehensibility messages differ, but also acceptance of the medium-comprehensibility message of Experiment 2 differed significantly from that of the good-comprehensibility message. Additional effects of lowered compre-

hensibility include more negative evaluations of the message and its source.

Interestingly, the effects of comprehension were general across levels of discrepancy and levels of counterarguing. Thus, even when the message took an extreme stand (i.e., two hours of sleep are desirable) and when subjects were instructed to counterargue, it was more persuasive to the extent that the supporting arguments were comprehensible.

Since no differences in perceived effort were obtained in either Experiment 1 or Experiment 2, effort justification (e.g., Zimbardo, 1965; Zimbardo & Ebbesen, 1970) is not an alternative hypothesis. However, subjects did report that the experience of listening to the poor-quality tape was less pleasant. Though unpleasantness may generally be inherent in an individual's not being able to understand a communication, the negative affect engendered by the difficult conditions of reception must be considered a possible mediator accounting for the lowered opinion change. In contrast, the viewpoint from which these investigations stemmed emphasizes the impact of reception processes on the amount of information an individual gains from the arguments supporting the conclusion of the message.

EXPERIMENT 3

In order to weigh the supportive argument versus affective explanations of the comprehensibility effects of Experiments 1 and 2 and to provide a baseline for assessing the effects of the differing levels of comprehensibility, Experiment 3 added a control condition in which the recommendation was supported by comprehensible but irrelevant arguments. This message, while not so unpleasant to receive, provided no argumentative support for the recommendation. In addition, to investigate further the generality of the relationship between comprehensibility and persuasion obtained in Experiments 1 and 2, this study employed a different comprehensibility manipulation. Experiment 3 used a 4 × 2 × 2 factorial design with four levels of comprehensibility (good, medium, poor, and irrelevant argument control), two levels of discrepancy size, and two levels of communicator credibility.

Method

Subjects

The subjects were 359 students (176 males and 183 females) enrolled in an introductory psychology course at the University of Massachusetts that offered extra-credit points for participation in experiments. One subject was eliminated for incomplete responding.

Procedure

Subjects were run in small groups. The experimenter explained that the overall purpose of the study was to obtain student reactions to lectures. She mentioned that the lectures would be presented in a written form but that the introductory page preceding the lecture would explain who wrote the lecture. She further explained that in order to take into account that students' comprehension of lectures varies, some subjects would be given lectures that would be difficult to understand. The method of construction of the various levels of comprehensibility (see below) was then briefly explained. This explanation made it impossible for subjects to attribute the disrupted ordering of the less comprehensible messages to the source's incompetence. Lectures varying in comprehensibility, discrepancy, and communicator credibility were then distributed, one to each subject. After 10 minutes, the lecture booklets were returned to the experimenter and the questionnaire administered. No-message control subjects responded only to the item concerning the desirable number of hours of sleep. The experiment was followed by debriefing.

Communicator Credibility

The communicator was described on a single sheet preceding the lecture. The high-credibility communicator was a physiological psychologist on the faculty of the Psychology Department at the University of Massachusetts "who has been doing research on sleep for the past five years. He has published a number of journal articles on sleep and is currently completing a book on the subject." The low-credibility communicator was described as a freshman psychology student who "researched the topic and wrote the lecture as part of a Psychology 101 project for which he had been assigned to construct an 'ideal lecture.'"

Persuasive Communication

The communication was a written version of the message on sleep used in Experiments 1 and 2. The two levels of discrepancy were six hours and one hour of sleep recommended, in contrast to the earlier studies, in which two hours was the most extreme discrepancy. Argument comprehensibility was manipulated by varying the ordering of the persuasive argument material. The arguments in the good-comprehensibility condition were an exact transcription from the message used in the earlier studies. In the medium-comprehensibility condition, the same

TABLE 3

RETENTION AND COMPREHENSION OF MESSAGE: EXPERIMENT III

Dependent variable	Good comprehensibility	Medium comprehensibility	Poor comprehensibility	Control: irrelevant arguments
Argument recall	3.62	2.85	1.32	.30
Recognition: multiple choice	11.77	10.09	8.16	4.17
Difficulty of comprehending[a]	6.00	2.54	1.11	3.72

Note. Within each dependent variable, all means are significantly different, $p < .05$, by the Tukey (*b*) test.
[a] Ratings are on a 7-point scale, with 7 indicating very easy and 1, very difficult.

sentences were presented but in a random order. In the poor-comprehensibility condition, the sentences were first cut in half, the first and last halves were separately randomized, and then they were joined (i.e., so that the first half of a sentence was joined with the last half of another sentence and both first and last halves appeared in a random order). In both the medium- and poor-comprehensibility conditions, half of the subjects received one random ordering, and half received a different ordering. In the irrelevant-argument condition, half of the subjects read a passage on ecology, while the other half read a passage on imprinting. These irrelevant messages were of the same length as those used in the other conditions.

Measuring Instruments

The postquestionnaire differed in a couple of respects from that used in Experiments 1 and 2. The message evaluation items were omitted. Instead, two new tasks involving responses to one-sentence versions of each of the persuasive arguments were introduced. In the first of these tasks, subjects indicated (on a 7-point scale extending from agree strongly to disagree strongly) how much they agreed with each of the arguments. When summed over the arguments, this measure correlated .42 with agreement with the communicator's overall recommendation. In the second task, subjects indicated (on a 7-point scale running from completely adequate justification to completely inadequate justification) how much justification each argument provided for sleeping only the number of hours per day that the lecturer recommended.

The two raters' counterargument scorings correlated .91, while their scorings of the number of arguments recalled correlated .96.

The two measures of acceptance of the communicator's recommendation correlated −.28, while the two retention measures correlated .65.

Results

Manipulation Checks

As the data in Table 3 indicate, comprehensibility again had a strong effect on retention of the message. Recall of the arguments improved as message comprehensibility increased ($F = 175.19$, $df = 3/316$, $p < .001$), as did recognition of the correct alternatives on the multiple-choice items ($F = 190.00$, $df = 3/316$, $p < .001$). Comprehensibility also had a highly significant impact on subjects' ratings of the difficulty of understanding the message ($F = 158.07$, $df = 3/316$, $p < .001$), though, not surprisingly, subjects in the irrelevant-argument condition rated the message as easier to understand than did subjects in the poor-comprehensibility condition.

Thirty-five subjects, who were retained in the analysis, failed to recall the communicator's recommendation accurately, though many of these subjects gave estimates only slightly different from the correct value. Somewhat surprisingly, inaccuracy was more common (17 out of 35) in the good-comprehensibility condition.

The professor communicator was rated as significantly more credible ($\bar{X} = 21.19$) than was the freshman student communicator ($\bar{X} = 18.27$, on a 7- to 35-point scale; $F = 15.36$, $df = 1/316$, $p < .001$). As the recommendation became more extreme, subjects reported less initial willingness to accept it ($p < .001$), and counterarguing increased ($p < .001$).

Acceptance of the Communicator's Recommendation

As shown in Table 4, agreement with the communicator's recommendation increased as comprehensibility improved ($F = 10.22$, $df = 3/316$, $p < .001$). By the Tukey (*b*) test, the good-comprehensibility mean differed significantly from each of the other means, which did not differ from one another. Agreement decreased as discrepancy size increased ($F = 52.32$, $df = 1/316$, $p < .001$). Com-

TABLE 4

AGREEMENT WITH THE COMMUNICATOR'S RECOMMENDATION: EXPERIMENT III

Discrepancy	Good comprehensibility	Medium comprehensibility	Poor comprehensibility	Control: irrelevant arguments
Small[a]	5.35 (43)	4.48 (44)	4.11 (36)	3.89 (44)
Large[b]	3.87 (43)	3.24 (42)	2.70 (37)	2.95 (43)
$\bar{X}$	4.59	3.87	3.40	3.42

Note. Ratings are on a 7-point scale, with 7 indicating strong agreement and 1, strong disagreement. Cell *n*s are given in parentheses.
[a] 6-hour recommendation.
[b] 1-hour recommendation.

municator credibility did not affect agreement ($F < 1$, $df = 1/316$), nor were there any significant interactions. Though the ideal number of hours of sleep estimated by the no-message control group subjects ($\bar{X} = 8.04$) was significantly different from that of the experimental group subjects ($\bar{X} = 7.09$; $t = 2.91$, $df = 357$, $p < .01$), none of the main effects or interactions were significant, though there were marginally significant tendencies for subjects to recommend fewer hours in response to the high-credibility communicator ($p < .06$) and the highly discrepant message ($p < .08$).

Table 5 shows that the ratings of the convincingness of the arguments increased strongly as comprehensibility improved ($F = 32.73$, $df = 3/316$, $p < .001$) but showed no other significant effects. Also, agreement with the arguments increased as message comprehensibility improved ($F = 7.64$, $df = 3/316$, $p < .001$). Credibility had a marginally significant effect on agreement with the arguments ($p < .07$), with greater agreement in the high-credibility condition. Table 5 also indicates that the extent to which the arguments were perceived as justifying the conclusion of the message also increased with comprehensibility ($F = 4.96$, $df = 3/316$, $p < .01$). In addition, a tendency for justification to lessen as discrepancy increased approached significance ($p < .06$).

Other Measures

As shown in Table 5, the source was rated more unfavorably as comprehensibility deteriorated ($F = 29.32$, $df = 3/316$, $p < .001$), and as reported above, the credibility manipulation had the expected effect on source ratings. No other source evaluation effects proved significant.

Subjects' ratings of the effort they put into reading the message, given in Table 5, were significantly affected by comprehensibility ($F = 3.70$, $df = 3/316$, $p < .05$), with the good-comprehensibility condition requiring less effort than did the other conditions. Table 5 also shows that the message was rated less pleasant as comprehensibility lessened ($F = 102.39$, $df = 3/316$, $p < .001$),

TABLE 5

DEPENDENT VARIABLES AS A FUNCTION OF COMPREHENSIBILITY: EXPERIMENT III

Dependent variable	Good comprehensibility	Medium comprehensibility	Poor comprehensibility	Control: irrelevant arguments
Convincingness of arguments[a]	4.70_x	3.64_y	2.56_z	2.46_z
Agreement with arguments[b]	32.24_x	30.94_x	28.93_y	28.70_y
Justification provided by arguments[b]	28.87_x	26.90_{xy}	25.96_y	25.22_y
Source evaluation[c]	25.00_x	20.22_y	16.75_z	16.47_z
Effort[a]	3.94_x	4.52_y	4.62_z	4.51_z
Pleasantness[a]	5.50_x	3.36_y	1.67_z	3.11_y
Distractingness[a]	2.48_x	5.79_y	6.77_z	5.49_y

Note. Within each dependent variable, means having a common subscript are not significantly different $p < .05$, by the Tukey (*b*) test.
[a] Ratings are on a 7-point scale, with 7 indicating maximum level of each attribute.
[b] Ratings range from 6 to 42, with 42 indicating maximum agreement or justification.
[c] Ratings range from 7 to 35, with 35 indicating most positive evaluation.

but as expected, the control message with the irrelevant arguments was not especially aversive. Ratings of the distractingness of the message, given in Table 5, showed a similar pattern ($F = 138.54$, $df = 3/316$, $p < .001$).

The number of statements written in response to the open-ended question used to elicit counterarguing was not affected by any of the experimental variables. However, the number of these statements that qualified as counterarguments was significantly affected by comprehensibility ($F = 3.03$, $df = 3/316$, $p < .05$), with the irrelevant-argument and medium-comprehensibility conditions eliciting more counterarguing than did the other two. However, this effect was complicated by the presence of a significant Comprehensibility × Credibility interaction ($F = 3.13$, $df = 3/316$, $p < .05$). Accordingly, counterarguing tended to increase as argument comprehensibility decreased in the low-credibility condition but tended to decrease in the high-credibility condition. Since this pattern is complex and is not mirrored in the agreement dependent variables, it is probably not important to the interpretation of the present study.

General Discussion

The manipulation of argument comprehensibility in Experiment 3 was based on disruptions of the ordering of sentences in a written version of the message rather than on the earlier procedure of introducing noise and distortion on a tape-recorded version. Yet the impact of the new manipulation on the retention of the content of the persuasive message and on the perceived difficulty of comprehending it was similar to that of the earlier studies, and the positive relationship between comprehensibility and agreement with the communicator's recommendation was again obtained. Thus, Experiment 3 established that this relationship is not dependent on a particular way of interfering with message reception.

It might be noted that the findings on one measure of acceptance of the message, how much sleep the subject thought that he should get, were weaker in Experiment 3. This measure was also used in Experiment 2, where it was again less sensitive than were other acceptance measures. These discrepancies, while troublesome, may be relatively unimportant because the number of hours of sleep index may be a somewhat indirect measure of agreement with the message: It concerns generalization of the recommendation to an individual's own life rather than agreement with the idea that a certain amount of sleep is in general desirable. This interpretation is substantiated by the tendency for the expected negative correlation between the two measures of message acceptance (i.e., high agreement was associated with a preference for a low number of hours of sleep) to be rather low in magnitude, though significant.

Variations in how discrepant the recommended position was from subjects' initial opinions had a consistent impact on opinion change in Experiments 1, 2, and 3. On the agreement measure, which did not remove the agreement differences between discrepancy conditions that existed prior to receipt of the message, acceptance of the message declined strongly as the recommended position became more extreme. In contrast, subjects' estimates of the number of hours of sleep that is desirable, a measure with a constant baseline across discrepancy conditions, did not relate systematically to discrepancy. The fact that this latter measure was not a positive function of discrepancy is inconsistent with most past research and especially with Bochner and Insko's (1966) study, which used the same issue.

Yet the more important consequence of the discrepancy variations is that they did not interact with comprehensibility in any of the three experiments. This finding is especially impressive in view of the even greater extremity of the large discrepancy recommendation in Experiment 3. Accordingly, improved comprehensibility of persuasive arguments brought a comparable increment of agreement for the recommendation to sleep only one hour per night, as it did for the six-hour recommendation.

Experiment 2 established the generality of the comprehensibility effects across levels of counterarguing. Furthermore, in Experiment 3, comprehensibility did not interact with communicator credibility (though the meaning of this finding might be questioned given the absence of significant main effects

of credibility on the agreement measures). In general, however, the lack of sensitivity of the argument comprehensibility effects to contextual cues is one of the strongly consistent features of this series of experiments. The phenomenon suggests an impartial message recipient who allows himself to be influenced by persuasive arguments to the extent that they are comprehensible, even though he may already have information (e.g., concerning how extreme a position the message takes) leading him to be quite unfavorably disposed to the overall position that the message advocates.

Mediation of Persuasion

Reception of persuasive arguments versus affective reactions. The main contribution of Experiment 3 is to understanding the psychological processes mediating the impact of message comprehensibility on acceptance of the message. Experiments 1 and 2 showed that both reception of persuasive arguments and affective reactions covaried with comprehensibility. Therefore, it was important to determine whether a person who receives a more comprehensible message is more accepting of its conclusion because of his clearer reception of the arguments supporting the conclusion or because of his lesser annoyance with the experience of listening to or reading the message.

Though independent manipulations of information reception and pleasantness were not possible, the inclusion of the irrelevant argument control condition in Experiment 3 did add useful information. One interesting comparison is between the control and poor-comprehensibility conditions. Since in the poor-comprehensibility condition subjects did comprehend some aspects of the arguments concerning sleeping, scores on the retention measures were significantly higher in this condition than in the control condition. However, the pleasantness ratings were significantly higher in the control condition, since this message, though irrelevant to the recommendation, did at least have a meaningful ordering of sentences. If the adequacy of subjects' reception of the arguments supporting the recommendation was the principal factor mediating the effect of comprehensibility on agreement, persuasion would be less in the control than in the poor-comprehensibility condition. If, on the other hand, pleasantness were crucial, persuasion would be greater in the control condition. The fact that no differences were obtained between these two conditions on either the agreement measures or the source evaluation ratings suggests that the amount of supportive information received and the affective reaction elicited by the reception situation both have a causal role. Thus, in the control condition, the increment of persuasiveness, due to its greater pleasantness, was countered by the decrement of persuasiveness, due to the absence of arguments in support of the recommendation.

The comparison between the medium-comprehensibility and the irrelevant-argument control conditions is also instructive since pleasantness was about equal in these conditions, while the amount of information held was considerably lower for the control subjects. Ratings of argument convincingness, agreement with the arguments, and source evaluation were significantly lower in the control condition. Agreement with the overall recommendation was lower, though not significantly so. Though there is some ambiguity created by the failure of this one comparison to reach significance, on an overall basis, these findings amount to fairly strong evidence of lower persuasion in the control compared to the medium-comprehensibility condition. This difference can be accounted for in terms of the lesser amount of supportive information received by the control subjects.

In addition, for all three experiments, analyses of covariance were of some value in comparing the supportive argument and affective interpretations of comprehensibility. Yet it must be kept in mind that such findings should be interpreted cautiously for two reasons: (*a*) As discussed above, the retention scores used in such analyses are a somewhat questionable index of reception, and (*b*) correlational relationships involving after-only agreement scores have the additional ambiguity of reflecting variance in initial position as well as response to the communication. The second problem is minimized somewhat (though not eliminated) in the

present experiment since initial opinions on the sleep issue have relatively low variance. At any rate, in all three experiments, covarying agreement scores on pleasantness or on either retention measure considerably diminished the strength of the argument comprehensibility effects, so that in most instances they were no longer significant. Also, retention scores and pleasantness ratings correlated positively with agreement in all three experiments. Correlations obtained when the data were pooled across experimental conditions were moderately strong, though average within-cell correlations tended to be lower. Thus, in general, the correlational findings did substantiate the interpretation that both the amount of supportive material received and the pleasantness of the conditions of reception contributed to the comprehensibility effects.

Some further attention should be given to the other dependent variables employed in these studies: ratings of the source and message, agreement with the arguments, and ratings of the degree to which each argument justified the overall conclusion. Not only did the relationship of comprehensibility to these additional dependent variables parallel those involving agreement with the conclusion, but also these variables were moderately strongly correlated with overall agreement. A possible interpretation of these dependent variables is that they function as mediators of persuasion. For example, it might be argued that lowering comprehensibility induced derogation of the source, which, in turn, lessened agreement with the conclusion. Or, as Fishbein and Ajzen (1972) would prefer to argue, lowering comprehensibility may lessen acceptance of the persuasive arguments and lower their perceived validity in relationship to the conclusion, and these reactions may, in turn, lessen acceptance of the conclusion. Whether any of these responses have such a mediating role or are merely additional manifestations of a more global reaction of acceptance versus rejection of the message cannot be determined from these data, and perhaps it is not a crucial issue. What is important about the present findings is that these measures do cohere and manifest a single pattern as a function of the comprehensibility manipulation. Interpretation of this overall pattern suggests that both the amount of supportive argument material received and pleasantness have a causal role in relationship to this entire set of responses.

The underlying processes involved in both the supportive argument and pleasantness effects deserve further discussion. However, the supportive argument effect seems fairly straightforward. By interfering with message reception, the comprehensibility manipulation lessened the amount of information that the subject held in support of the conclusion, and as a consequence, agreement with the conclusion lessened.

Explaining why messages that are more pleasant to receive are more persuasive is more problematic. There are a number of possible interpretations. For example, Staats's (e.g., Staats, 1968; Staats & Staats, 1958) notions concerning the classical conditioning of attitudes are relevant if we assume that it is the pairing of the pleasant or unpleasant conditions of reception with the message that is crucial. Or, the message might also be thought of as conveying information about the conditions of reception (e.g., "This message is unpleasant to hear") in addition to the argumentative information. The information concerning pleasantness then functions as an additional determinant of persuasion. Clearly, further research concerning the impact of pleasant and unpleasant stimuli on persuasion is needed to clarify the issue. The "eating while reading" studies (Dabbs & Janis, 1965; Janis, Kaye, & Kirschner, 1965) provide one example of research into problems of this type.

Finally, interpretation of the contrast between the control and the poor-comprehensibility conditions in terms of pleasantness and amount of supportive information received must be taken as somewhat tentative. Other factors, not rated by the subjects, could conceivably have differed between these two conditions and affected persuasion. In addition, though retention of the message was significantly greater in the poor-comprehensibility condition than in the control condition, it is possible that the argumentative material received in the poor-comprehensibility condition was nevertheless too minimal to produce an increment in the persuasiveness of the

message. This logic would weaken the inference that pleasantness mediated agreement. Though this interpretation cannot be ruled out completely, it seems somewhat less plausible, since comparable increments in message reception between the other comprehensibility conditions did affect agreement with the message.

Counterarguing and other mediators. The present experiments also yielded information concerning the mediating role of counterarguing. The fact that the counterarguing manipulation in Experiment 2 affected agreement with the message confirms the generalization that counterarguing lessens persuasion. This manipulation, which was based on instructing subjects to counterargue, is perhaps more direct and clear-cut than those that involve providing subjects with information, as in McGuire's refutational defense pretreatments (e.g., McGuire & Papageorgis, 1961). In addition, correlations between counterarguing and opinion change were consistently negative.

While the notion that heightened counterarguing lessens persuasion was, indeed, substantiated, it should be kept in mind that the message comprehensibility effects were not mediated by counterarguing. However, counterarguing did covary with discrepancy, a finding suggesting that the lesser agreement with the extreme recommendation may be accounted for by the greater tendency of subjects to counterargue it.

Other possible mediating processes also did not appear important in explaining the impact of argument comprehensibility on opinion change. Effort related to comprehensibility only in Experiment 3, but lower effort occurred in the good-comprehensibility condition, a finding that runs counter to the effort justification hypothesis (e.g., Zimbardo & Ebbesen, 1970). Nor is the interpretation of the present findings in terms of distraction helpful, even though the perceived distractingness of the situation varied with message comprehensibility. Research has shown that distraction is a global variable that is a function of at least two mediating processes: (*a*) Under some conditions, distraction increases persuasion by lowering the level of counterarguing (e.g., Osterhouse & Brock, 1970); (*b*) under other conditions, distraction inhibits persuasion by interfering with message reception (e.g., Haaland & Venkatesan, 1968).

Sex Differences

Since all three experiments used approximately equal numbers of male and female subjects, the research provided an excellent opportunity to evaluate sex differences in persuasibility as well as the psychological processes that might mediate any such effects. Though past research has sometimes found females more persuasible than males (e.g., Janis & Field, 1959; King, 1959), such a finding was not obtained in any of the present studies. However, in Experiment 3, interactions involving sex reflected a tendency for females to agree less with the highly discrepant message, especially when it was comprehensible. Sex differences in message retention are also of interest in view of the possibility that the greater influenceability of women, when it occurs, may be due to their superior comprehension of verbal materials. Females were better able to recall the arguments used in the message in Experiment 1. A trend ($p < .08$) in the same direction occurred in Experiment 2, but there were no significant differences in Experiment 3. No effects of sex were obtained on the recognition measure of message retention.

Implications for an Information-Processing Model of Persuasion

The implications of the present research for an information-processing approach to understanding persuasion are moderately optimistic. The traditional importance given to message reception processes (e.g., Hovland et al., 1953) is supported: Interfering with the reception of persuasive arguments does, indeed, have a strong and consistent effect on acceptance of the message. Yet the research also suggests that the mechanism through which this relationship is likely to function is not solely a product of the amount of argumentative material received. The subject's negative affective reaction to attempting to comprehend information that is difficult to understand appears to be important also.

However, the long-range issue of principal concern in attitude-change research is not whether interfering with the comprehensibility of a message *can* lessen acceptance of a message. Rather, the issue is whether naturally occurring variations of situational and individual difference factors interfere with message comprehensibility so as to lower the persuasiveness of communications. The manipulations of comprehensibility used in the present research were, after all, very strong, and it may not be surprising that they affected opinion change.

A number of questions should be posed in future research. For example, do variations in situational and dispositional variables make enough of a difference in message comprehensibility that opinion change is significantly affected? If opinion change is affected, would the mechanisms involve the supportive information and pleasantness factors suggested by the present analysis? McGuire's (1968) analysis of the relationship of personality variables to influenceability argued in favor of the mediating role of reception processes. However, empirical support for his particular approach to the issue has been rather ambiguous (e.g., Millman, 1968; Nisbett & Gordon, 1967; Zellner, 1970).

In summary, the viewpoint emerging from the present series of experiments is in one respect simpler and in another respect more complex than what was originally anticipated. The simplicity arises from the finding that comprehensibility does not interact with other variables affecting the level of agreement with the communicator's recommendation. The complexity arises in specifying the psychological processes responsible for the impact of comprehensibility on persuasion. Since both supportive information and affective reactions appear to be important, this series of experiments highlights the need to integrate knowledge of affective and motivational factors with our growing understanding of the information-processing mechanisms that mediate persuasion.

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